

Algorithm generateSubsets(set, currentSubset, index)		
1	if index == length(set),	1
2	print(currentSubset)	1
3	return	1
	// Case 1 Exclude the current element and move to the next	
4	generateSubsets(set, currentSubset, index + 1)	$T(n-1)$
	// Case 2 Include the current element and move to the next	
5	generateSubsets(set, currentSubset + set[index], index + 1)	$T(n-1)$

$$T(n) = 1 + 1 + 1 + T(n-1) + T(n-1)$$

$$= 3 + 2T(n-1)$$

using c for representing constants

$$T(n) = 2T(n-1) + c$$

original

$$T(n) = 2T(n-1) + c$$

↓

substitute $T(n-1) = 2T(n-2) + c$

$$= 2 \cdot [2 \cdot T(n-2) + c] + c$$

$$= 2^2 T(n-2) + 2c + c$$

$$T(n) = 2^2 T(n-2) + 3c$$

↓

$$T(n-2) = 2T(n-3) + c$$

$$= 2^2 [2T(n-3) + c] + 3c$$

$$= 2^3 T(n-3) + 2^2 c + 3c$$

$$= 2^3 T(n-3) + 4c + 3c$$

$$T(n) = 2^3 T(n-3) + 7c$$

through recursion it created a pattern

$$T(n) = 2^k T(n-k) + c$$

substituting $T(n-1)$

$$T(n) = 2T(n-1) + c$$

$$T(n-1) = 2T(n-1-1) + c$$

$$T(n-1) = 2T(n-2) + c$$

substituting $T(n-2)$

$$T(n-2) = 2T(n-2-1) + c$$

$$T(n-2) = 2T(n-3) + c$$